INSIDE STUDIO SP · VST3

ALIGN

KICK / BASS PHASE

Step-by-step manual

Version 0.2.0 · Windows and macOS · VST3 and AU · Free



insidestudiosp.com.br/align · contato@insidestudiosp.com.br

What ALIGN is

ALIGN lines up the **phase of the kick and the bass**, with **a single button**. It was built for the psytrance rolling bass (K b B B), where the kick tail lands on top of the first bass note and the two cancel out (or turn to mud) in the low end.

- **Continuous automatic correction.** On every hit it measures how the kick tail and the bass note sum in the low end and picks, out of 34 combinations of polarity and phase rotation, the one that sums without cancelling.
- **Per-note memory (NOTE MEMORY).** Stores the correction for each bass note and only switches if it improves by more than 1 dB. The switch is smooth (10 ms, while the kick is sounding).
- **LEVEL GUARD.** Aligning must not turn into a level boost: where kick and bass overlap, the bass low end is pulled down only by whatever would exceed the power sum (no hole, no mud), down to −6 dB.
- **Bass low end only.** The X-OVER follows the note (about 1.6 × the fundamental). The rest of the bass passes through as is.

Before you start

ITEM	DETAILS
Format	VST3 and AU, Windows and macOS.
Where it goes	On the **BASS** channel, with the **KICK** via sidechain.
Latency	**Zero.** No look-ahead: safe to use while recording or live.
Switched off	Audio passes through completely untouched (handy for A/B), and ALIGN keeps listening and learning.

Download and activation (free)

1. Go to **insidestudiosp.com.br/align**, enter your e-mail and choose Windows or Mac.
2. The **token** and the download link appear on screen. Keep the token.
3. Run the installer and open your DAW (if needed, have it scan for new plugins).
4. The first time, enter the **e-mail** and the **token** (ISP-XXXX-…).

> **LICENSE.** The free ALIGN license works on 1 computer. Internet is only needed for activation; after that it re-checks itself every 45 days. Lost your token? The ALIGN page has a "perdi meu token" (I lost my token) option that sends it by e-mail. To see the license: `≡` menu › `Licença…` (License…).

> **WITHOUT ACTIVATION**, the plugin opens but passes audio straight through.

STEP 1

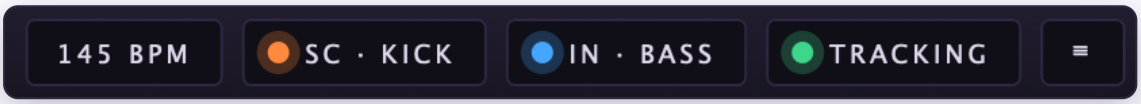
1 Put ALIGN on the BASS

Insert ALIGN on the **bass** channel. If you use BAREBASS, put ALIGN **after** it.

STEP 2

2 Send the KICK via sidechain

In ALIGN's **Sidechain**, select the **KICK** track. In Ableton: open the plugin's *Sidechain* section, select the kick track and set it to **Post FX**.



Top of the window: BPM, SC · KICK (sidechain), IN · BASS (input), the state and the ≡ menu.

STATE AT THE TOP	MEANING
NO SIDECHAIN	The kick isn't coming in. Check the sidechain.
BYPASS	ALIGN button off: audio passes through untouched (but it keeps listening).
LEARNING	On and learning the notes.
TRACKING	On and correcting hit by hit.

STEP 3

3 Hit play and see the problem

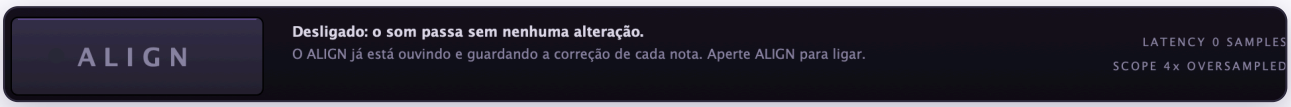
With the track playing, the oscilloscope locks to every kick hit and **LOW END** shows how much low end is disappearing or building up where kick and bass overlap:

- **LOST TO PHASE CANCELLATION**: kick and bass are cancelling each other (low end disappearing).
- **BUILD-UP (KICK + BASS SUMMING)**: the two are summing too much (getting muddy).
- **IN PHASE**: all good.

STEP 4

4 Press ALIGN

Click the big `ALIGN` button at the bottom left. That's it: it corrects on its own, hit by hit. The text next to the button explains what's happening.



Off: audio passes through untouched, and ALIGN is already listening and storing the correction for each note.



On: aligns the phase and holds the level. Notes in purple in NOTE MEMORY already have a stored correction.

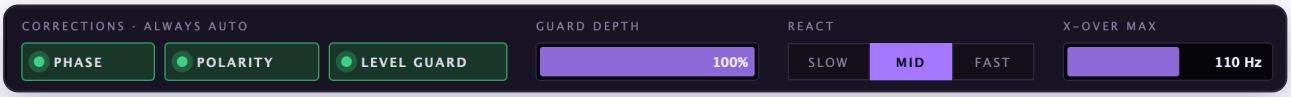
A / B. Toggle the ALIGN button on and off to compare. Off is exactly the original sound.

FIRST HIT. Since latency is zero, the first hit of a note it has never heard may come out with the previous note's correction. From the second hit on, it's right.

STEP 5

5 Dial in the correction (optional)

The correction is always automatic (CORRECTIONS · ALWAYS AUTO). The strip below the oscilloscope just sets how much of each part is applied:



CONTROL	RANGE / DEFAULT	WHAT IT DOES
PHASE	on/off · **on**	Phase rotation of the bass low end.
POLARITY	on/off · **on**	Polarity inversion. With PHASE and POLARITY off, the low end passes straight through.
LEVEL GUARD	on/off · **on**	Level control where kick and bass overlap.
GUARD DEPTH	0–100% · **100%**	How much LEVEL GUARD pulls down.
REACT	SLOW / **MID** / FAST	How fast it learns each note and switches correction.
X-OVER MAX	60–150 Hz · **110 Hz**	Upper limit of the corrected low end. The X-OVER keeps following the note below this limit.

On the sliders (GUARD DEPTH and X-OVER MAX): click or drag; **double-click** returns to the default.

Reading the screen



OSCILLOSCOPE · 2 BEATS

Kick (orange) and bass (blue) over 2 beats, locked to the kick. `FULL` shows the full signal; `LOW < X Hz` shows only the low end (below the X-OVER). The K, b and B marks show the kick and the bass notes, and the dashed box marks where they overlap: **IN PHASE** or **OUT OF PHASE**. The drawing is 4x (interpolated) and doesn't touch the audio.

ZOOM · KICK → b · SUM

The handover from the kick tail to the first bass note (b): the sum (**SUM**, white), the target = power sum (**TARGET**, purple dashed) and, with ALIGN on, the sum you would get without LEVEL GUARD (pink).

LOW END

dB lost (cancellation) or added where they overlap. With ALIGN on, it should sit near 0 dB, showing **NO CANCEL · NO BUILD-UP** and the current note.

CORRELATION · LOW

From −1 to +1: how closely the kick and bass low end move together. Near +1 is good; negative means cancellation.

CORRECTION (AUTO)

POLARITY (NORMAL / INVERT), **ROTATION** (degrees), **X-OVER** (Hz) and **LOW DUCK** (how much LEVEL GUARD pulls down). Any part that's switched off shows OFF.

NOTE MEMORY

In purple, the notes with a stored correction. White outline = the current note.

FAQ and troubleshooting

SITUATION	WHAT TO DO
NO SIDECHAIN / "Sem kick no sidechain" (no kick on sidechain) shows up	Select the kick track in ALIGN's Sidechain (in Ableton: Post FX).
The sound didn't change	Check that the ALIGN button is on (LEARNING or TRACKING state) and that the license is active (≡ › Licença…). If LOW END was already near 0, there was nothing to fix.
The bass low end got quieter	That's LEVEL GUARD holding back the sum. Lower GUARD DEPTH or turn LEVEL GUARD off.
It's correcting the bass mids	Lower X-OVER MAX.
Switches correction too often / too rarely	Use REACT: SLOW to switch less, FAST to react faster.
The first hit of a new note sounds different	Normal with zero latency: from the second hit on, it's right.
I want to compare	Turn off the ALIGN button: the sound goes back to exactly the original.

30-second summary

1. ALIGN on the **BASS** channel.
2. **KICK** on the sidechain (Ableton: Post FX).
3. Play: watch **LOW END**.
4. Press **ALIGN**.
5. If you want, dial in with PHASE, POLARITY, LEVEL GUARD, GUARD DEPTH, REACT and X-OVER MAX.

> **SUPPORT.** Questions, suggestions or problems: **contato@insidestudiosp.com.br** · **insidestudiosp.com.br**